

Experiment - 07

Student Name: Kamal Sharma
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70380
Section/Group: 24AIT_KRG-1_B
Date of Performance: 31-08-26
Subject Code: 24CSP-310

Question 1

Problem:

Consider a relation $R(A, B, C, D)$ with functional dependencies:

$$AB \rightarrow C, \quad C \rightarrow D, \quad D \rightarrow A$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attribute B never appears on the right-hand side (RHS) of any dependency, so B must be included in every candidate key.
- Computing attribute closures:
 - $(B)^+ = \{B\}$
 - $(AB)^+ = \{A, B, C, D\} \rightarrow AB$ is a candidate key
 - $(BC)^+ = \{B, C, D, A\} \rightarrow BC$ is a candidate key
 - $(BD)^+ = \{B, D, A, C\} \rightarrow BD$ is a candidate key
- **Candidate Keys:** $\{AB, BC, BD\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B, C, D\}$
- **Non-Prime Attributes:** $\emptyset$ (None)

3. Normal Form Analysis:

- **BCNF Test:** For $C \rightarrow D$, C is not a superkey. Thus, R violates BCNF.
- **3NF Test:** A dependency $X \rightarrow Y$ satisfies 3NF if X is a superkey OR

Y is a prime attribute. Since all attributes in R are prime attributes, every dependency satisfies 3NF.

- **Highest Normal Form: 3NF**

Question 2

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$A \rightarrow D, \quad B \rightarrow A, \quad BC \rightarrow D, \quad AC \rightarrow BE$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attribute C never appears on the RHS, so C must be part of every candidate key.
- Computing attribute closures:
 - $(C)^+ = \{C\}$
 - $(AC)^+ = \{A, C, B, E, D\} \rightarrow AC$ is a candidate key
 - $(BC)^+ = \{B, C, A, D, E\} \rightarrow BC$ is a candidate key
 - $(CD)^+ = \{C, D\}$
 - $(CE)^+ = \{C, E\}$
- **Candidate Keys: $\{AC, BC\}$**

2. Attribute Classification:

- **Prime Attributes: $\{A, B, C\}$**
- **Non-Prime Attributes: $\{D, E\}$**

3. Normal Form Analysis:

- **2NF Test:** A relation is in 2NF if no non-prime attribute is partially dependent on any candidate key. In $A \rightarrow D$, A is a proper subset of candidate key AC , and D is a non-prime attribute (partial dependency).
- **Highest Normal Form: 1NF**

Question 3

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$B \rightarrow A, \quad A \rightarrow C, \quad BC \rightarrow D, \quad AC \rightarrow BE$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Computing attribute closures:
 - $(A)^+ = \{A, C, B, E, D\} \rightarrow A$ is a candidate key
 - $(B)^+ = \{B, A, C, E, D\} \rightarrow B$ is a candidate key
- **Candidate Keys:** $\{A, B\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B\}$
- **Non-Prime Attributes:** $\{C, D, E\}$

3. Normal Form Analysis:

- In each FD ($B \rightarrow A, A \rightarrow C, BC \rightarrow D, AC \rightarrow BE$), the determinant (LHS) contains either A or B , making every LHS a superkey.
- **Highest Normal Form: BCNF**

Question 4

Problem:

Consider a relation $R(A, B, C, D, E, F)$ with functional dependencies:

$$A \rightarrow BCD, \quad BC \rightarrow DE, \quad B \rightarrow D, \quad D \rightarrow A$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attribute F never appears on the RHS, so F must be part of every

candidate key.

- Computing attribute closures:
 - $(F)^+ = \{F\}$
 - $(AF)^+ = \{A, F, B, C, D, E\} \rightarrow AF$ is a candidate key
 - $(BF)^+ = \{B, F, D, A, C, E\} \rightarrow BF$ is a candidate key
 - $(DF)^+ = \{D, F, A, B, C, E\} \rightarrow DF$ is a candidate key
 - $(CF)^+ = \{C, F\}$
 - $(EF)^+ = \{E, F\}$
- **Candidate Keys:** $\{AF, BF, DF\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B, D, F\}$
- **Non-Prime Attributes:** $\{C, E\}$

3. Normal Form Analysis:

- In $A \rightarrow BCD$, $A \rightarrow C$ maps a proper subset of candidate key AF to a non-prime attribute C (partial dependency).
- **Highest Normal Form: 1NF**

Question 5

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$CE \rightarrow D, \quad D \rightarrow B, \quad C \rightarrow A$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attributes C and E never appear on the RHS, so CE must be part of every candidate key.
- $(CE)^+ = \{C, E, D, B, A\}$
- **Candidate Keys:** $\{CE\}$

2. Attribute Classification:

- **Prime Attributes:** $\{C, E\}$
- **Non-Prime Attributes:** $\{A, B, D\}$

3. Normal Form Analysis:

- In $C \rightarrow A$, C is a proper subset of candidate key CE , and A is a non-prime attribute (partial dependency).
- **Highest Normal Form: 1NF**

Question 6

Problem:

Consider a relation $R(A, B, C, D, E, F)$ with functional dependencies:

$$AB \rightarrow C, \quad DC \rightarrow AE, \quad E \rightarrow F$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attributes B and D never appear on the RHS, so BD must be part of every candidate key.
- $(BD)^+ = \{B, D\}$
- $(ABD)^+ = \{A, B, D, C, E, F\} \rightarrow ABD$ is a candidate key
- $(BCD)^+ = \{B, C, D, A, E, F\} \rightarrow BCD$ is a candidate key
- **Candidate Keys:** $\{ABD, BCD\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B, C, D\}$
- **Non-Prime Attributes:** $\{E, F\}$

3. Normal Form Analysis:

- In $DC \rightarrow AE$, $DC \rightarrow E$ has LHS DC which is a proper subset of candidate key BCD , and E is non-prime (partial dependency).
- **Highest Normal Form: 1NF**

Question 7

Problem:

Consider a relation $R(A, B, C, D, E, F, G, H, I)$ with functional dependencies:

$$AB \rightarrow C, \quad BD \rightarrow EF, \quad AD \rightarrow GH, \quad A \rightarrow I$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attributes A, B , and D never appear on the RHS, so ABD must be part of every candidate key.
- $(ABD)^+ = \{A, B, D, C, E, F, G, H, I\}$
- Candidate Keys:** $\{ABD\}$

2. Attribute Classification:

- Prime Attributes:** $\{A, B, D\}$
- Non-Prime Attributes:** $\{C, E, F, G, H, I\}$

3. Normal Form Analysis:

- In $AB \rightarrow C$, AB is a proper subset of candidate key ABD , and C is non-prime (partial dependency).
- Highest Normal Form: 1NF**

Question 8

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$AB \rightarrow CD, \quad D \rightarrow A, \quad BC \rightarrow DE$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attribute B never appears on the RHS, so B must be part of every

candidate key.

- $(B)^+ = \{B\}$
- $(AB)^+ = \{A, B, C, D, E\} \rightarrow AB$ is a candidate key
- $(BD)^+ = \{B, D, A, C, E\} \rightarrow BD$ is a candidate key
- $(BC)^+ = \{B, C, D, E, A\} \rightarrow BC$ is a candidate key
- **Candidate Keys:** $\{AB, BD, BC\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B, C, D\}$
- **Non-Prime Attributes:** $\{E\}$

3. Normal Form Analysis:

- **BCNF Test:** In $D \rightarrow A$, D is not a superkey, violating BCNF.
- **3NF Test:** For $D \rightarrow A$, while D is not a superkey, the RHS attribute A is a prime attribute. All other dependencies have superkeys on their LHS ($AB \rightarrow CD, BC \rightarrow DE$).
- **Highest Normal Form: 3NF**

Question 9

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$BC \rightarrow ADE, \quad D \rightarrow B$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- Attribute C never appears on the RHS, so C must be part of every candidate key.
- $(BC)^+ = \{B, C, A, D, E\} \rightarrow BC$ is a candidate key
- $(CD)^+ = \{C, D, B, A, E\} \rightarrow CD$ is a candidate key
- **Candidate Keys:** $\{BC, CD\}$

2. Attribute Classification:

- **Prime Attributes:** $\{B, C, D\}$
- **Non-Prime Attributes:** $\{A, E\}$

3. Normal Form Analysis:

- **BCNF Test:** In $D \rightarrow B$, D is not a superkey, violating BCNF.
- **3NF Test:** In $D \rightarrow B$, B is a prime attribute. In $BC \rightarrow ADE$, BC is a superkey.
- **Highest Normal Form: 3NF**

Question 10

Problem:

Consider a relation $R(A, B, C)$ with functional dependencies:

$$A \rightarrow B, \quad B \rightarrow C, \quad C \rightarrow A$$

Identify the candidate keys, prime attributes, non-prime attributes, and the highest normal form.

Solution:

1. Candidate Keys:

- $(A)^+ = \{A, B, C\}$
- $(B)^+ = \{B, C, A\}$
- $(C)^+ = \{C, A, B\}$
- **Candidate Keys:** $\{A, B, C\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, B, C\}$
- **Non-Prime Attributes:** $\emptyset$ (None)

3. Normal Form Analysis:

- Every determinant (A, B, C) is a candidate key and therefore a superkey.
- **Highest Normal Form: BCNF**

Question 11

Problem:

Consider a relation $P(F_1, F_2, F_3, F_4, F_5)$ with functional dependencies:

$$F_1 \rightarrow F_3, \quad F_2 \rightarrow F_4, \quad (F_1, F_2) \rightarrow F_5$$

Identify the normal form and describe the prime and non-prime attributes.

Solution:**1. Candidate Keys:**

- F_1 and F_2 never appear on the RHS, so (F_1, F_2) must be part of every candidate key.
- $(F_1, F_2)^+ = \{F_1, F_2, F_3, F_4, F_5\}$
- **Candidate Key:** $\{F_1 F_2\}$

2. Attribute Classification:

- **Prime Attributes:** $\{F_1, F_2\}$
- **Non-Prime Attributes:** $\{F_3, F_4, F_5\}$

3. Normal Form Analysis:

- In $F_1 \rightarrow F_3$, F_1 is a proper subset of candidate key $F_1 F_2$ and F_3 is non-prime (partial dependency).
- **Highest Normal Form: 1NF**

Question 12**Problem:**

Consider a relation $R(A, B, C, D, E, F, G, H, I, J)$ with functional dependencies:

$$AB \rightarrow C, \quad AD \rightarrow GH, \quad BD \rightarrow EF, \quad A \rightarrow I, \quad H \rightarrow J$$

Identify the normal form and candidate keys, decompose to 3NF, and state the number of decomposed tables.

Solution:**1. Candidate Keys:**

- Attributes A, B , and D never appear on the RHS, so ABD must be part of every candidate key.
- $(ABD)^+ = \{A, B, C, D, E, F, G, H, I, J\}$

- **Candidate Key:** $\{ABD\}$
- 2. **Attribute Classification:**
 - **Prime Attributes:** $\{A, B, D\}$
 - **Non-Prime Attributes:** $\{C, E, F, G, H, I, J\}$
- 3. **Normal Form Analysis:**
 - $AB \rightarrow C$, $AD \rightarrow GH$, $BD \rightarrow EF$, and $A \rightarrow I$ all involve proper subsets of the candidate key determining non-prime attributes (partial dependencies).
 - **Highest Normal Form: 1NF**
- 4. **Decomposition into 3NF:**
 - Decompose partial dependencies:
 - $R_1(A, B, C)$ for $AB \rightarrow C$
 - $R_2(A, D, G, H)$ for $AD \rightarrow GH$
 - $R_3(B, D, E, F)$ for $BD \rightarrow EF$
 - $R_4(A, I)$ for $A \rightarrow I$
 - Retain the candidate key:
 - $R_5(A, B, D)$
 - Decompose transitive dependency:
 - $R_6(H, J)$ for $H \rightarrow J$ (with H remaining in R_2 as a foreign key)
 - **Total Tables Formed: 6 tables**

Question 13

Problem:

Consider a relation $R(A, B, C, D, E)$ with functional dependencies:

$$A \rightarrow B, \quad B \rightarrow E, \quad C \rightarrow D$$

Identify the normal form and candidate keys, decompose to 3NF, and state the number of decomposed tables.

Solution:

1. **Candidate Keys:**

- Attributes A and C never appear on the RHS, so AC must be part of every candidate key.
- $(AC)^+ = \{A, B, C, D, E\}$
- **Candidate Key:** $\{AC\}$

2. Attribute Classification:

- **Prime Attributes:** $\{A, C\}$
- **Non-Prime Attributes:** $\{B, D, E\}$

3. Normal Form Analysis:

- In $A \rightarrow B$, A is a proper subset of candidate key AC and B is non-prime (partial dependency).
- **Highest Normal Form: 1NF**

4. Decomposition into 3NF:

- Create relations for each functional dependency:
 - $R_1(A, B)$ for $A \rightarrow B$
 - $R_2(B, E)$ for $B \rightarrow E$
 - $R_3(C, D)$ for $C \rightarrow D$
- Ensure candidate key preservation:
 - $R_4(A, C)$
- **Total Tables Formed: 4 tables**